CECS 451
Term Project (11 Weeks)
Total: 150 Points

General Instructions

- Submit all deliverables to the Canvas Dropbox (no email submissions).
- Team size: **3–4 students**.
- Project window: **Sep 21–Dec 11, 2026**.
- LLM usage is **encouraged but not mandatory**.
- Every submission must identify the team members and the project title. One team member may submit on behalf of the team unless otherwise announced.
- Do not include passwords, API keys, private data, or other secrets in the report, video, or GitHub repository.

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1. **Project Objective**
   - Design and prototype an **AI-powered business innovation** (product, service, or platform) that solves a real-world problem or creates new value.
   - Emphasis: **AI application**, **business modeling**, and **team-based project execution**.
   - The project must demonstrate a working AI-enabled user flow. A slide-only concept, static mockup, or unverifiable claim is not sufficient for full credit.
   - The team deliverables are worth 100 points. An additional 50 points are awarded individually through the contribution and peer-review process, for a maximum of **150 points**.
2. **Timeline & Phases – Fall 2026 (All phase deadlines are Fridays @ 23:59 unless otherwise specified)**

   Phase 0 – Team Formation (0 points)
   Deliverable: **Team-member Canvas Groups registration**
   Due: Sep 25, 2026 (Fri)
   - Recruit the required 3–4 team members and finalize the team composition.
   - Submit the complete team-member list through the Canvas `People - Groups` menu.
   - Phase 0 is an administrative checkpoint and is worth **0 points**. It is required before the team begins the graded project work.

Phase 1 – Ideation, Market Validation & Feasibility (Weeks 1–4)

Deliverable: **Integrated Ideation, Market Validation & Feasibility Report (maximum 4 pages, excluding references)**
Due: Oct 16, 2026 (Fri)

- **Industry and target user:** Identify the industry, target customer/user, and the specific situation in which the problem occurs.
- **Problem definition:** Describe the customer pain point, its current consequences, and why the problem is important.
- **Proposed solution:** Explain the product, service, or platform and the main user workflow.
- **AI application plan:** Clearly state the AI task, input, output, proposed model/API/algorithm, and why AI is appropriate instead of a rule-based or manual solution.
- **Initial scope:** Define the minimum feature set and the intended user flow for the prototype.
- **Market and existing solutions:** Summarize relevant customers, competitors, comparable products, or existing approaches and identify the gap your project addresses. Cite the sources used.
- **Business feasibility:** Explain the value proposition, expected users/customers, possible revenue or cost-saving model, and major adoption barriers.
- **Technical approach:** Select appropriate AI techniques (LLM, NLP, CV, recommendation, etc.) and explain why the selected approach fits the problem.
- **Data plan:** Identify the data source, access method, license/permission, privacy considerations, preprocessing, and a realistic evaluation approach. If data is unavailable, describe a realistic fallback plan.
- **Risks and milestones:** Identify important data, privacy, bias, safety, business, or feasibility risks and list the specific tasks needed to reach a working PoC and MVP.
- The team may revise or pivot the initial idea after market validation. A change in industry, problem, target user, or solution is allowed when the feasibility analysis supports the change and is not penalized by itself.

Phase 1 checklist: A reader should be able to understand the problem, target user, proposed AI solution, market gap, business feasibility, technical approach, data plan, risks, milestones, and final project direction.

Phase 2 – Prototype Development (Weeks 5–9)

Deliverables:

- **Technical documentation in the GitHub repository** (architecture, setup, and code explanation)
- **Prototype Progress Report (maximum 2 pages, must include the GitHub repository URL)**

Due: Nov 20, 2026 (Fri)

- The progress report must include: project objective and current scope; completed features; system architecture or workflow diagram; data and model/algorithm used; preliminary output or results; known limitations; remaining tasks; and each member's current responsibility.
- Build a **PoC** to verify core technical feasibility and extend it toward an **MVP** with a working end-to-end user flow.
- Examples include a chatbot, recommender, vision-based analysis tool, or dashboard. Other project types are acceptable if they demonstrate a meaningful AI-enabled workflow.
- Implement at least one AI model or algorithm. If an external LLM or AI API is used, document the prompts/API workflow, input-output behavior, limitations, and the team's own technical contribution.
- The repository documentation must explain how to install dependencies, configure the system, run the prototype, reproduce a sample result, and understand the main files. Use environment variables or another safe method for credentials.

Phase 2 checklist: A grader should be able to run or inspect the prototype, understand the architecture, see what was implemented, and distinguish completed work from planned work.

Phase 3 – Evaluation & Final Presentation (Weeks 10–11)

Deliverables:

- **Pitch Presentation Video (10 minutes maximum)** (must include a prototype demo; upload to YouTube or a similar platform – do not upload video files to Canvas; **verify that the video is accessible in incognito/private browsing mode before submission**)
- **Combined Final Report (maximum 8 pages, must include the Pitch Presentation Video URL link)**

Due: Dec 11, 2026 (Fri)

- The final report is a single document that combines the revised Phase 1 and Phase 2 reports. It is not a separate new report with a new set of requirements.
- The Phase 1 and Phase 2 reports may simply be merged as submitted. Editing the combined report for consistency and readability is allowed but not required. Do not repeat cover pages or duplicate the same material across phases.
- The video must cover: problem, target users, solution, AI technology, working prototype demo, evidence of results, business applicability, limitations, and roadmap.
- The prototype demo must be part of the submitted video and must show an actual input-to-output workflow. Screenshots alone do not constitute a demo.
- Performance evaluation and business-impact evidence are assessed primarily through the Phase 3 presentation video, prototype demo, and GitHub repository.

Phase 3 checklist: A grader should be able to verify what was built, how well it works, what evidence supports the claims, what business value it creates, and what each team member contributed.

3. **Evaluation Rubric (150 points)** The Phase 1, Phase 2, and Phase 3 team deliverables are worth 100 points. The additional Individual Contribution and Peer Review component is worth 50 points per student, for a maximum of 150 points.

Phase	Team Points	Evaluation Items
Phase 1 – Ideation, Market Validation & Feasibility	40 points	**Problem and target user** (6) **Business importance and value** (5) **AI application plan** (6) **Scope and risks** (3) **Market research and gap** (5) **Business feasibility** (4) **AI approach and technical feasibility** (5) **Data plan and responsible use** (4) **Risks, milestones, and clarity** (2)
Phase 2 – Prototype	30 points	**Technical implementation and depth** (9) **AI/model/LLM integration** (8) **Working PoC-to-MVP progression** (7) **Documentation and reproducibility** (4) **Creativity and design decisions** (2)
Phase 3 – Evaluation & Pitch	30 points	**Performance evaluation and analysis** (12) **Business impact and scalability** (8) **Prototype demo and technical communication** (6) **Presentation organization and professionalism** (4)

How the rubric is applied

- Each Phase 1, Phase 2, and Phase 3 evaluation item is scored independently using the points shown above. The item scores within each phase add exactly to that phase's weight, and the three scored phases add to 100 team points.
- The additional 50 individual points are determined by the Individual Contribution and Peer Review process described below. The final project score is the team score (0–100) plus the individual score (0–50), for a maximum of 150 points.
- **Full credit:** The required element is complete, specific, supported by evidence where appropriate, and consistent with the project scope.
- **Partial credit:** The element is present but incomplete, insufficiently specific, weakly supported, or not fully connected to the proposed AI/business problem.
- **Little or no credit:** The element is missing, unverifiable, copied without meaningful project contribution, or inconsistent with the submitted prototype and evidence.
- Requirements and grading criteria will be clarified before the relevant deadline. New requirements will not be applied retroactively to work submitted before the clarification.

4. **Individual Contribution and Peer Review (50 points)** The 100 team points are shared by the team. The additional 50 points are assigned individually to prevent free riding and to recognize participation across the entire project.

   Submission timing and format

   - Each student submits one confidential peer review by the Phase 3 deadline.
   - The review must consider the student's participation throughout **Phase 1, Phase 2, and Phase 3**; it is not a Phase-specific review.
   - The review must list each other team member's name and assign one overall participation rating from **0–2**. Self-ratings are not submitted.
   - Students may optionally attach evidence that supports a rating, but evidence is not required.

   Rating scale

   - **2 – Full/Substantial participation:** consistently made substantial, reliable, and active contributions throughout the project.
   - **1 – Partial/Moderate participation:** completed a reasonable share of work, but with notable inconsistency or less contribution than expected.
   - **0 – Minimal or no participation:** made little or no meaningful contribution to the project (free riding).

   Scoring method

   - For each student, the instructor calculates the mean of the ratings received from the other team members. Self-ratings are not included.
   - The individual contribution score is:

$$\text{Individual Contribution Points} = 50 \times \frac{\text{mean peer rating}}{2}.$$

   - A verified free rider (mean rating of 0) receives 0 individual points.

   Instructor moderation and free-rider protection

   - Peer reviews are confidential to the instructor. A single unusually low or high rating is not used automatically to penalize or reward a student.
   - If ratings are clearly inconsistent, missing, retaliatory, or collusive, the instructor may review project records already submitted by the team, such as GitHub history, report revisions, and prototype or presentation participation. Students may also optionally submit supporting evidence with the peer review.
   - When necessary, the instructor may adjust the calculated individual score to reflect the documented level of participation. A student with verified nonparticipation may receive 0 individual points.

- Failure to submit the required peer review by the Phase 3 deadline results in a provisional 0 individual contribution score, unless the student has an approved reason.

5. **Late Submission Policy**

- All deliverables must be submitted by the deadline unless otherwise specified below.
- **Phases 1–2:**
  - Late submission within **1 week** after the deadline is accepted with a **20% penalty**.
  - Submissions more than 1 week late will **not be accepted**.
- **Phase 3 (Final Report & Presentation):**
  - Late submission is accepted with **no penalty** until **Dec 15, 12:15 PM**.
  - Submissions after **Dec 15, 12:15 PM** will **not be accepted**.

6. **Notes on Terms**

- **PoC (Proof of Concept)**: Minimal experiment to verify technical feasibility of the core idea.
- **MVP (Minimum Viable Product)**: Prototype with minimum features to demonstrate value and gather feedback.
- **AI application plan**: A concise description of the target user, AI task, input, output, proposed AI method, data, initial scope, and major risks.
- **Prototype Progress Report**: A short report describing the current implementation, architecture, preliminary evidence, limitations, remaining work, and team responsibilities. It is not a second proposal.
- **Technical documentation**: Repository documentation that enables a reader to understand and, when reasonably possible, run the prototype. It should include setup, dependencies, usage, architecture, and known limitations.
- **Performance evaluation**: Evidence showing how well the AI system performs. It may include task-appropriate quantitative metrics, human evaluation, representative examples, comparison with a baseline, and failure analysis.
- **Pitch presentation**: A concise presentation of the problem, solution, AI method, evidence, business value, and roadmap; **the 10-minute video must include an actual prototype demo**.